

Controller Step-Response Comparison Report

Lap: 2 | Generated: 2026-04-13 17:35:01

Metrics: Mean Rise Time, Overshoot, Settling Time (X and Y axes), plus tracking error context

Interpretation notes:

- Not Settled: response did not enter and remain within the $\pm 2\%$ settling band in the evaluation window.
- Not Available: metric could not be computed (e.g., transition too short or no valid crossing).

Executive Summary

Best overall mean step-response score: Pure Pursuit (score=1.67)

Fastest mean Y rise time: NN AI Controller (0.928 s)

Lowest mean Y overshoot: Pure Pursuit (3.615 %)

Step-Response Metrics

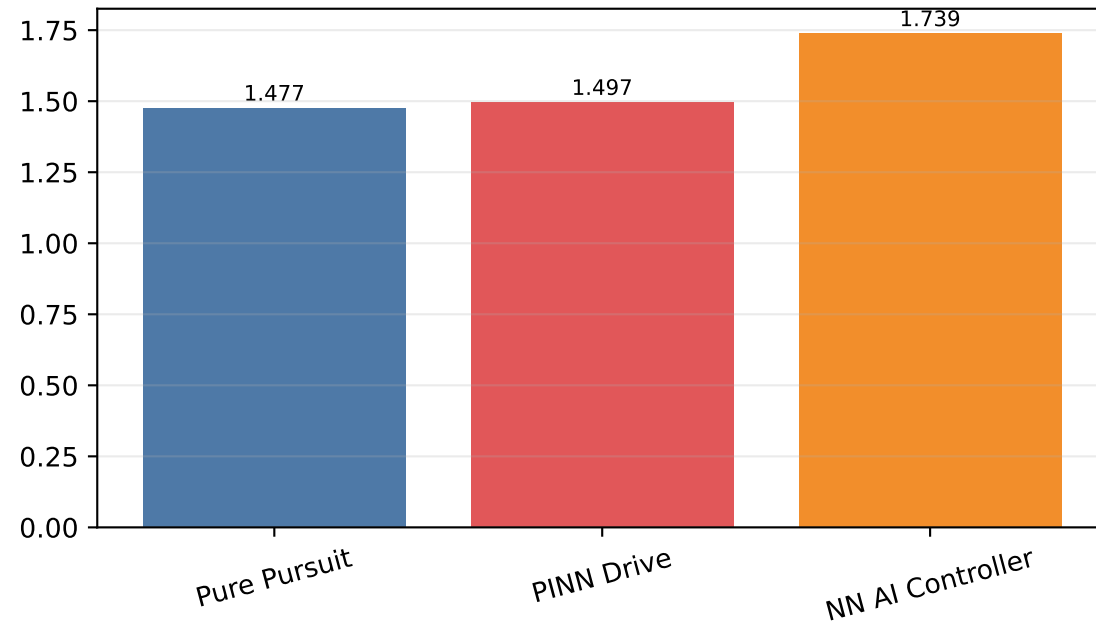
Controller	X Rise [s]	Y Rise [s]	X Overshoot [%]	Y Overshoot [%]	X Settling [s]	Y Settling [s]	Score
Pure Pursuit	1.477	1.189	1.751	3.615	0.960	1.669	1.667
PINN Drive	1.497	0.954	2.959	4.617	Not Settled	1.323	2.400
NN AI Controller	1.739	0.928	2.920	5.601	Not Settled	1.578	2.600

Tracking Error Context

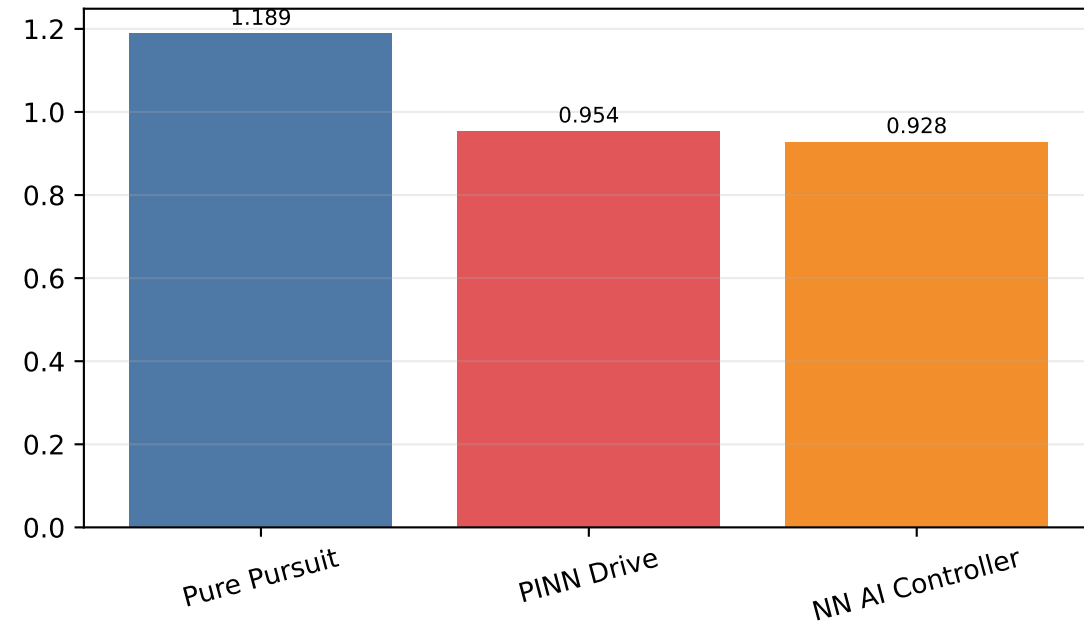
Controller	Lap Time [s]	RMSE [m]	Mean Error [m]	Max Error [m]
Pure Pursuit	9.003	2.592	2.369	4.213
PINN Drive	8.151	1.401	1.155	2.969
NN AI Controller	8.805	1.703	1.589	3.026

Mean Step-Response Metrics by Controller

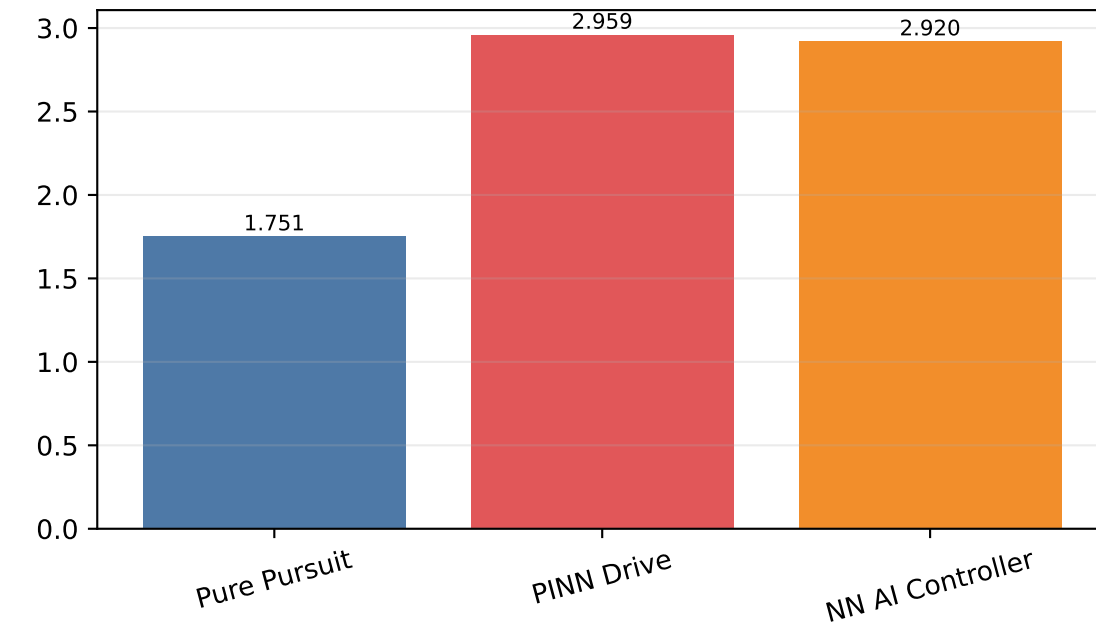
Mean Rise Time X [s]



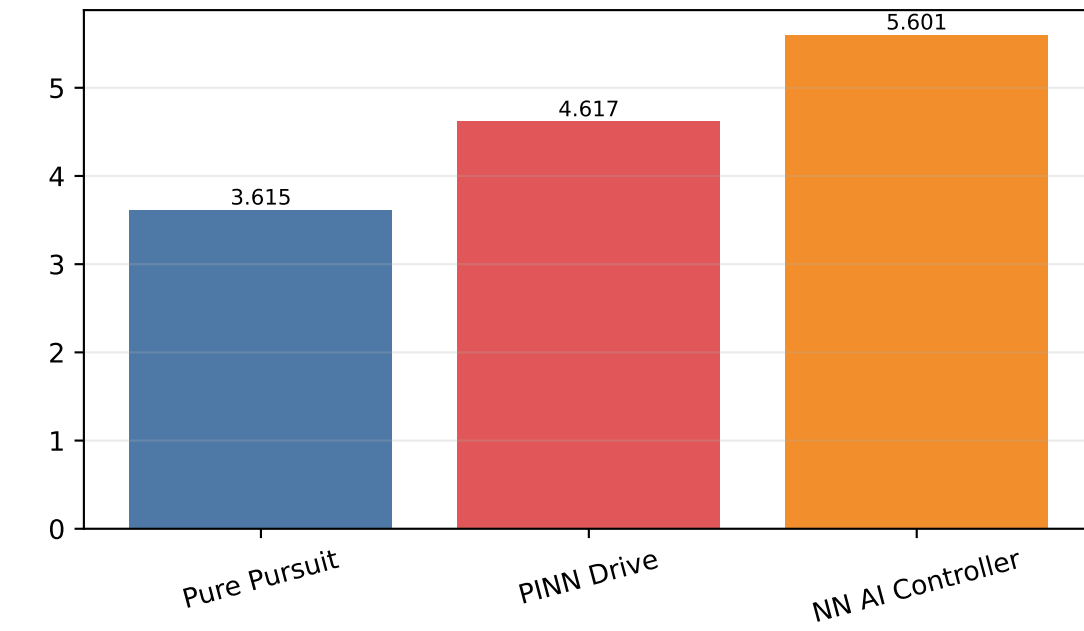
Mean Rise Time Y [s]



Mean Overshoot X [%]

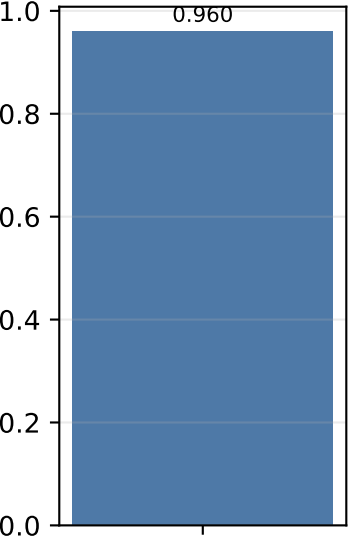


Mean Overshoot Y [%]



Settling Performance and Transition Coverage

Mean Settling Time X [s]

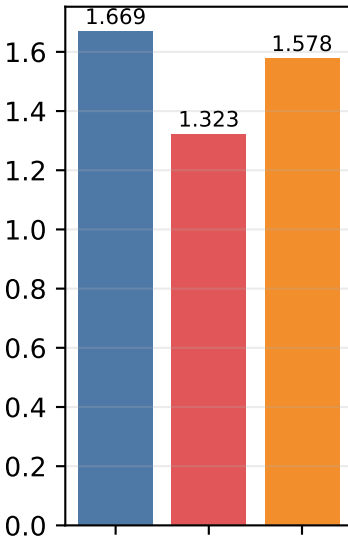


Pure Pursuit

Not Settled

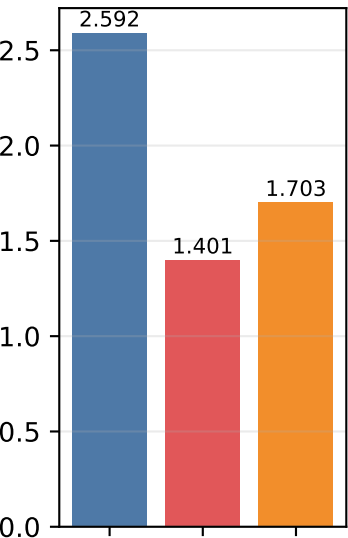
Not Settled

Mean Settling Time Y [s]



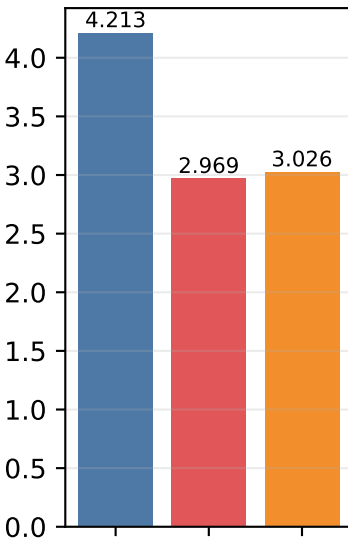
Pure Pursuit
PINN Drive
NN AI Controller

Position RMSE [m]



Pure Pursuit
PINN Drive
NN AI Controller

Max Position Error [m]



Pure Pursuit
PINN Drive
NN AI Controller

Transition-Level Step-Response Details (Lap 2)

Note: 'Not Settled' means the response did not settle in the evaluated transition window.

Controller	Axis	Transition	Rise [s]	Overshoot [%]	Settling [s]	Steady-State Error
Pure Pursuit	x	1	0.681	2.954	0.960	0.278
Pure Pursuit	x	2	2.227	2.299	Not Settled	8.953
Pure Pursuit	x	3	1.522	0.000	Not Settled	-1.267
Pure Pursuit	y	1	1.275	3.711	1.745	0.297
Pure Pursuit	y	2	1.103	3.520	1.593	0.027
PINN Drive	x	1	0.729	6.641	Not Settled	-0.564
PINN Drive	x	2	2.235	2.238	Not Settled	9.599
PINN Drive	x	3	1.526	0.000	Not Settled	-1.096
PINN Drive	y	1	0.868	5.164	1.446	-0.196
PINN Drive	y	2	1.040	4.069	1.199	0.268
NN AI Controller	x	1	0.924	6.293	Not Settled	-0.775
NN AI Controller	x	2	2.287	2.466	Not Settled	9.216
NN AI Controller	x	3	2.004	0.000	Not Settled	-1.257
NN AI Controller	y	1	0.952	5.477	1.578	-0.244
NN AI Controller	y	2	0.904	5.724	Not Settled	0.506

Appendix: Lap-2 Tracking Comparison (Time Domain)

Tracking Comparison | Lap 2 | X / Y / CTE vs Time

